Hence potential difference across thermistor (voltmeter reading) increases.

23 Voltage is constant ➔ $V = IR$

$$I = \frac{V}{R} = \frac{V}{\left(\frac{\rho L}{A}\right)} = \frac{VA}{\rho L} = \frac{VA}{\rho(2\pi r)N}$$

where A is cross sectional area of wire and L is length of wire.

Magnetic flux density remains constant.

B of flat circular coil = $\frac{\mu_0 NI}{2r} = \frac{\mu_0 N}{2r}\left(\frac{VA}{\rho(2\pi r)N}\right)$

= $\frac{\mu_0 VA}{4\pi r^2 \rho}$

Therefore, B is independent of N. It will only stay constant if r remains the same in this case.